**3.** **In this question you must show all stages of your working.**

**Solutions relying entirely on calculator technology are not acceptable.**

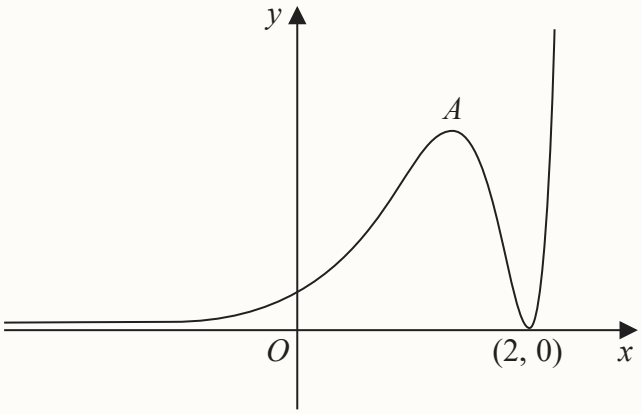



**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = (x - 2)^2 \mathrm{e}^{3x} \qquad x \in \mathbb{R}$$

The curve has a maximum turning point at $A$ and a minimum turning point at $(2, 0)$

(a) Use calculus to find the exact coordinates of $A$.

**(5)**

Given that the equation $\mathrm{f}(x) = k$, where $k$ is a constant, has **at least** two distinct roots,

(b) state the range of possible values for $k$.

**(2)**